Semester II Examinations 2018/2019

Exam Code(s)	1SD3, 1SD1, 1MF1
Exam(s)	Higher Diploma in Software Design and Development MSc in Software Design and Development
Module Code(s)	CT537
Module(s)	Software Engineering Methods
Paper No.	1
Repeat Paper	
External Examiner(s)	Prof. Pier Luca Lanzi
Internal Examiner(s)	Prof. Michael Madden Ms. Karen Young*

Instructions: Candidates should answer **any 3** questions.

Duration	2 hours
No. of Pages	5
Discipline(s)	Information Technology
Course Co-ordinator(s)	Dr. Enda Barrett, Dr. Seamus Hill

Requirements:

Release in Exam Venue	Yes	X	No	
MCQ	Yes		No	X
Handout	None			
Statistical/ Log Tables	None			
Cambridge Tables	None			
Graph Paper	None			
Log Graph Paper	None			
Other Materials	None			
Graphic material in colour	Yes		No	x

1.(a) Construct a **Class Diagram**, illustrating the relevant **domain classes** (attributes and operations), and their **relationships** (associations, inheritance) based on the textual description of an *online golf club shop* presented below:

> John Arnold, a golf shop owner, and golf professional, has decided to offer a customised golf club shop on the Internet. During his ten years in business, he has had many enquiries about purchasing individual golfclubs (usually sold in sets) and about extending or shortening the club shaft length, without incurring the prohibitive costs of tailor-made clubs. In order to reach a sufficient audience (broadly distributed), he wants to set up a Web site so that customers can browse his stocks, enter orders and receive goods.
>
> The main aim of this system is to support the buying, marketing, selling and distribution of golf clubs, and the management of financial accounts. After interviewing John Arnold, you note the following requirements:
>
> - Customers will need to register with Arnold GolfStyle to make orders. On registration, they need to provide their name and address, payment details, golf club requirements and any other special details.
> - To order, customers will select from the golf club range. The system will tell them if the clubs are in stock and how quickly they can be delivered or if they have to be ordered from a supplier.
> - Customers will want to track their orders online. An order can be waiting for delivery to Arnold GolfStyle, waiting for dispatch, waiting for credit clearance or dispatched. Before dispatch, the customer should have an option for cancelling the order.
> - The system has to support stock management and John does not want to hold large quantities of stock himself as this will only increase his warehouse storage costs. When goods are ordered, the stock levels should be deducted and checked to see if the stock has hit reorder level.
> - John wants a weekly report detailing the numbers of customers, orders, stock and cancelled orders.
> - Registered customers will receive monthly promotional literature outlining any special offers.

(15)

(b) Assume you are leading the project to develop the *online golf club shop* described above. Provide a brief report detailing how the **implementation** and **testing** of the system will be approached. The report should detail the design method, programming language and framework employed, stating all assumptions made.

(5)

2. (a) Prepare a **Use Case Diagram** for the *Rent-a-Car* Car Rental system described below, and expand ONE of the use cases to a full textual description.

> The car rental agency, *Rent-a-car*, provides a range of vehicles for rental to customers. To rent a vehicle a customer must complete a rental request with their name, address, date of birth, and telephone number. They must also provide a current driving licence, passport, current insurance policy number and proof of
> place of residence. These details are recorded in the system in order to create a customer account. The customer's details are recorded on a rental contract, which is signed by the customer. One copy of this contract is retained by the Rent-a-car agency, and another is given to the customer.
>
> To hire a car, customers can browse the agency's catalogue and select the appropriate category (ranging from A1 to F5 in order of vehicle cost and size) for their needs. The employee then validates the relevant customer's details (driving licence and insurance details) and ensures that the customer has no outstanding account with the company. If these conditions are satisfied, the employee enters the details of the rental transaction in the system. The customer pays for the car-hire, including a damage deposit, and is given the car keys along with a receipt and their copy of the rental contract.
>
> When the customer returns a car, it is inspected by a mechanic. Fines are deducted from the customer's deposit before the deposit is returned to the customer. If the deposit is insufficient to pay the fines, these additional charges must be paid by the customer (charged to their credit card) before the account can be closed. The keys are retrieved from the customer, and returned to the key hanger behind the counter.
>
> *Rent-a-car* requires that an outstanding charges report be generated weekly. This lists all fines accruing from late returns, damaged vehicles or insurance claims. This report is used to issue reminders to customers. If charges are not paid within three weeks from the due date, the customer is recorded as having defaulted and their account is forwarded to a credit collection agency.

(10)

(b) Prepare an *Activity* diagram for this *Rent-a-Car* Car Rental system using swimlanes as required to illustrate the different processes.

(10)

3. (a) Prepare a **State Model** (**State Transition Diagram**) for the coffee making machine described below:

> The coffee maker consists of four buttons (A, B, C and D). The coffee maker can grind coffee beans into powder and it can then brew the coffee using this powder. The use of the grinder is optional as the user may just add coffee powder directly and disable the grinder.
>
> Button A controls the power on/off to the coffee maker. Therefore, if the coffee maker is powered on and A is pressed then it powers down. If the coffee maker is currently powered down and A is pressed it will then power up.
>
> Button B controls the grinder on/off function. When initially powered up the grinder is enabled. If B is pressed it disables the grinder. If B is pressed again it enables the grinder and so on.
>
> Button C controls the hotpan on/off function. The coffee pot is placed on the hotpan when brewing the coffee. If the hotpan is enabled it will generate heat for 60 minutes keeping the coffee warm. When the coffee maker is initially powered up the hotpan functionality is enabled. It can be disabled by pressing the button C. Pressing C again will re-enable the hotpan.
>
> Button D controls the on/off brewing process. When D is pressed the machine begins to brew the coffee (it may or may not have to first grind the beans). The brewing process will only commence if there is water supplied in the coffee maker (the user adds the water) and if the lid of the coffee machine is fully closed. While pressing D will enable the brewing process, pressing D again will suspend the brewing process.

(10)

(b) How would you define software **quality**? Briefly evaluate the contribution of THREE software engineering techniques to improving the quality of software produced.

(5)

(c) You have been appointed a **software project manager** for a company that services the genetic engineering world. Your job is to manage the development of a new software product that will accelerate the pace of gene typing. The work is R&D oriented, but the goal is to produce a product within the next year. List five major **risks** of this project and a **mitigation strategy** for each.

(5)

4. (a) "*Furious activity is no substitute for understanding*" H.H. Williams

What **software engineering practices** do you believe support the developer in gaining an understanding of the problem domain? In your experience, are these practices adequate? What other practices do you think could help with this challenging phase of the SDLC? Give reasons to support your answer.

(10)

(b) You are the team lead of a systems design team that is implementing a multiple module project with an expected duration of eighteen months. Your client just sent you the memo included below. Write a memo back to your client answering their questions.

> *Dear Lead,*
>
> *I was reading **IT Update** magazine and an article mentioned that many project teams now use **prototyping** and **agile** techniques to speed up their systems development processes. What are these techniques? Will they save us time and money? What problems might we encounter if we use them?*
>
> *Regards,*
> *XYZ.*

(6)

(c) Compare and contrast the various **behavioural diagrams in UML** (what they model, when and why they are used) illustrating your answer with examples to clarify their differences.

(4)